

Low-Resource Neural Machine Translation Using Recurrent Neural Networks and Transfer Learning: A Case Study on English-to-Igbo

OCHEME ANTHONY EKLE^{*†‡}, Tennessee Technological University, USA

BISWARUP DAS^{*†}, Moscow Institute of Physics and Technology, Russia

In this study, we developed a Neural Machine Translation (NMT) model for English-to-Igbo translation—targeting a low-resource African language spoken by over 40 million people across Nigeria and West Africa. We trained our model using a curated and benchmarked dataset compiled from Bible corpora, local news, Wikipedia articles, and Common Crawl, all verified by native language experts. Our approach utilizes Recurrent Neural Network (RNN)-based architectures, including Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), enhanced with attention mechanisms to improve translation accuracy. To further boost performance in this low-resource setting, we applied transfer learning using the MarianNMT pre-trained models within the SimpleTransformers framework. Our RNN-based system achieved competitive results, closely matching existing English-Igbo benchmarks. With transfer learning, we achieved a performance gain of +4.83 BLEU points, reaching an estimated translation accuracy of 70%. These results highlight the effectiveness of combining RNNs and transfer learning to bridge the performance gap in low-resource language translation tasks.

CCS Concepts: • **Computing methodologies** → **Machine translation**; **Natural language processing**.

Additional Key Words and Phrases: Neural Machine Translation, Low-Resource Languages, English-to-Igbo Translation, RNN, LSTM, Natural Language Processing, Transfer learning

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1 INTRODUCTION

The need for advanced translation systems and the increasing diversity of languages have raised several challenges in translating from one language to another. This has led to significant research in the fields of Natural Language Processing (NLP) and machine translation [2]. Machine Translation (MT) is one of the major applications of NLP, employing automated systems to translate text between languages. MT takes a sentence in a source language and translates it into a corresponding sentence in a target language. It is essential that the meaning of both the source and target sentences remains the same [65].

Past NLP researchers have adopted a variety of approaches to build machine translation systems. These include Rule-Based Machine Translation (RBMT), Corpus-Based or Statistical Machine Translation (SMT), and Hybrid Machine

^{*}Both authors contributed equally to this research.

[†]This research was conducted while the author was at **Moscow Institute of Physics and Technology (MIPT)**, Russia.

[‡]Author is currently affiliated with Tennessee Technological University, USA.

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Translation (HMT), which combines RBMT and SMT techniques. These approaches encompass word-based, phrase-based, forest-based, and Neural Machine Translation (NMT) methods [54].

Neural Machine Translation (NMT) represents a significant advancement in the field, where a deep neural network is trained to read and translate text [65]. NMT typically employs an encoder-decoder architecture, consisting of an encoder block that processes the source sentence and a decoder block that generates the target sentence [18]. NMT architectures have demonstrated impressive results compared to previous translation techniques and have achieved near state-of-the-art performance [3].

Studies have shown that NMT systems learn faster and produce more accurate translations than traditional techniques. Recurrent Neural Network (RNN)-based models are commonly used in NMT due to their ability to capture sequential data effectively. However, despite the success of NMT, little attention has been given to capturing endangered languages with limited datasets, particularly African languages [26]. Hence, this research focuses on developing an NMT model to translate English into the Igbo language—an African language spoken by over 40 million people across Nigeria and West Africa.

The key contributions of our work are summarized as follows:

- We present the first well-performing translation system for the low-resource English-Igbo language pair, incorporating the attention mechanism [3] and global attention [47].
- We integrated the teacher forcing algorithm to provide accurate reference information to the decoder layer, thereby improving the translation quality.
- We implemented both Greedy Decoding and Beam Search techniques to enhance the performance of our model.
- We conducted detailed experiments on approximately 12,000 low-resource parallel corpora. Our model demonstrated improved translation quality beyond existing benchmarks. Results from our transfer learning approach [51, 53, 78] achieved an optimal translation accuracy of 70% and a notable improvement of +4.83 BLEU points over the existing state-of-the-art (SOTA).

The remainder of this paper is structured as follows: Section 2 reviews relevant literature in Natural Language Processing (NLP) and Neural Machine Translation (NMT). Section 3 introduces the foundational concepts of Recurrent Neural Networks (RNNs) and discusses their inherent limitations. In Section 4, we describe the architecture of our proposed translation model. Section 5 details the experimental setup, including comparisons of RNN variants, attention mechanisms, decoding strategies, datasets, and the application of transfer learning. Section 8 presents the quantitative and qualitative results, along with performance evaluations. Finally, Sections 8 and 9 provide a comprehensive discussion, conclusion, identified limitations, and directions for future research.

2 RELATED WORK

This section provides a review of the literature on Natural Language Processing (NLP), levels of language understanding, word representation, and machine translation. It also highlights recent developments in graph-based learning and generative models, showing how they relate to low-resource language translation.

2.1 Natural Language Processing and Language Levels

Research in NLP began in the late 1940s, combining concepts from artificial intelligence (AI) and linguistics, with machine translation (MT) as one of its earliest applications [45, 50]. In 1947, Warren Weaver proposed using cryptography and information theory for language translation [63]; however, his approach struggled with lexical ambiguity. This

challenge inspired deeper exploration in the field. Noam Chomsky’s landmark 1957 work, *Syntactic Structures*, introduced generative grammar, significantly shaping NLP and MT [44]. This era also saw early efforts in speech recognition, heuristic reasoning, and question answering. Between the 1970s and 1990s, NLP advanced with innovations like semantic parsing and discourse modeling [41]. Schank and Tesler developed language understanding programs focused on human conceptual knowledge, such as memory and goals [61].

Since the 1990s, probabilistic and data-driven methods have dominated NLP, with models like probabilistic parsing and word embeddings. Advances in computing hardware have further enhanced speech and language processing capabilities [45]. Today, researchers focus on next-generation applications that better handle linguistic variability and ambiguity. A core objective is to design computational techniques that learn and represent text data in a human-like manner. Converting symbolic text into numerical form remains a foundational step in NLP.

Language Levels: NLP operates across multiple linguistic levels: phonology, morphology, lexical, syntactic, semantic, pragmatic, and discourse [41]. These layers are interdependent and essential for understanding and generating language. Ambiguity remains a major challenge, addressed using knowledge-based methods (e.g., Mahesh and Nirenburg, 1996) and modern techniques like FastText [67]. **Phonology** deals with sound patterns; **morphology** analyzes word structure; **syntax** examines sentence formation; **lexicon** interprets word meanings; **semantics** studies meaning within phrases; **discourse** handles coherence across text units; and **pragmatics** evaluates intent and contextual meaning [45].

2.2 Word Vector Representation

Word vector representations are crucial in NLP for capturing the semantic relationships between words [12]. Early methods used “one-hot” encodings, which lacked the ability to capture similarity between words. Firth’s idea of representing words by their context laid the foundation for embeddings. Bengio et al. [6] introduced neural word embeddings, followed by Collobert and Weston [16], who applied multitask and semi-supervised learning.

Mikolov et al. [49] introduced Word2Vec, featuring the continuous bag-of-words (CBOW) and skip-gram architectures. CBOW predicts a word given its context, while skip-gram predicts context from a given word. GloVe, proposed by Pennington et al. [56], is another widely used pretrained embedding. Facebook’s FastText [9] captures subword information and polysemy, and later work incorporated uncertainty modeling [1].

More recently, contextualized word embeddings such as BERT [21], ELMo, and GPT-2 [25] have surpassed older models, outperforming both one-hot and bag-of-words methods [31]. These embeddings are now central to both supervised and unsupervised tasks, powering neural machine translation, question answering, and information retrieval.

In this study, we employ the Keras tokenizer to convert text into numerical sequences using morphology-based tokenization. We also use pretrained word embeddings via Keras’ embedding layer to generate dense vector representations.

2.3 Neural Machine Translation

The origin of MT dates back to Warren Weaver’s 1946 proposal of using cryptographic techniques for language translation [36]. In later years, researchers began combining rule-based and statistical methods, giving rise to hybrid models for neural machine translation (NMT) [73].

Modern NMT systems typically follow the sequence-to-sequence (Seq2Seq) paradigm, composed of an encoder and decoder [65]. The encoder transforms the input sentence into a fixed-length vector (hidden state), while the decoder generates the output sequence based on this vector. However, traditional encoder-decoder models face two primary challenges:

(1) Long-distance dependencies between words can degrade translation accuracy. Gusev and Oboturov addressed this by reversing the encoder direction to assist training. (2) Compressing the entire input sequence into a single vector introduces an **information bottleneck**, especially for long inputs.

Bahdanau et al. [3] addressed these challenges using an attention mechanism. Rather than relying on a single fixed vector, attention allows the decoder to reference all encoder states dynamically at each step, focusing on relevant input tokens during decoding. This greatly improved translation quality, but the method increased computational complexity and still struggled with low-resource data and long-term dependencies.

Luong et al. [47] later introduced local attention, improving alignment between source and target tokens. Self-attention networks, such as those developed by Yang et al. [74] and Shaw et al. [64], further advanced sequence modeling by enabling tokens to attend to different positions within a sentence.

Besides RNN-based models like LSTM and GRU, CNNs have also been explored for NMT. Kalchbrenner et al. [38] and Bradbury et al. [10] applied CNNs for efficient sequence modeling, while Gehring et al. [27] used CNNs with attention to model long dependencies. Other innovations include Connectionist Temporal Classification (CTC) by Graves et al. [32] and model ensembling for robust predictions [52].

A breakthrough in NMT came with the Transformer architecture proposed by Vaswani et al. [68]. This model replaced recurrence with self-attention, enabling parallelization and improved performance. Key components include multi-head attention and position-wise feed-forward networks. Transformers laid the foundation for models like GPT-2 [25], BERT [33], and T5 [58].

Despite their success, Transformer-based models often underperform on low-resource languages [78]. To address this, transfer learning has been proposed: a high-resource (parent) model is trained first, and its knowledge is transferred to a low-resource (child) model [51, 53, 78]. This approach improves performance by leveraging shared representations.

2.4 Graph Neural Networks and Emerging Trends in NLP

While traditional NLP models rely on sequential or convolutional architectures, recent research has shown that Graph Neural Networks (GNNs) [4] can effectively model structured relationships in linguistic data, such as syntactic and semantic dependencies [69]. GNNs have been applied in tasks like word embedding [22], knowledge graph completion, and low-resource machine translation by capturing long-range interactions across words and phrases in a graph structure [22, 46]. For instance, in translation tasks, representing words as nodes and their co-occurrence or syntactic relations as edges enables the model to preserve richer context.

In parallel research, GNN-based techniques have also proven effective for anomaly detection in dynamic graph streams [7, 13, 23, 24]. These contributions highlight the growing role of GNNs in both structured NLP and dynamic graph-based learning.

Furthermore, the emergence of Large Language Models (LLMs), such as GPT, multilingual BERT variants, and recent generative embedding techniques, has pushed the boundaries of translation and multilingual understanding. These models leverage large-scale pretraining to capture context-rich representations and enable robust performance in low-resource settings [11, 72]. Generative embeddings are increasingly being used in cross-lingual retrieval, summarization, and neural translation [60, 70], offering new opportunities for integrating contextual, graph-based, and generative learning.

In this research, we focus on English-to-Igbo translation—a low-resource language pair. We adopt the RNN-based encoder-decoder framework and enhance it with attention and global attention mechanisms [3, 47]. We also apply

the teacher forcing algorithm [42] to guide decoding during training and incorporate transfer learning to optimize performance.

Goal: By leveraging these techniques, we aim to build an effective neural machine translation model that bridges the language gap between English and Igbo, contributing to the development of low-resource NLP applications.

3 BACKGROUND AND RNN ARCHITECTURES

This section provides the foundational background for Recurrent Neural Networks (RNNs), including the classical RNN architecture and its challenges. We also explore advanced variants such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU). Furthermore, we introduce the sequence-to-sequence (Seq2Seq) encoder-decoder model, highlight its limitations, and describe how the attention mechanism addresses those challenges.

3.1 Traditional RNNs and the Vanishing Gradient Problem

Recurrent Neural Networks (RNNs) are designed to model temporal or sequential data by maintaining hidden states that capture information from previous time steps. This ability makes them suitable for a variety of time-dependent tasks, including financial forecasting [30], electric load prediction [17], and water quality monitoring [48].

Early RNN architectures, such as the Vanilla RNN, Elman Network, and Jordan Network [57], encountered limitations due to the vanishing and exploding gradient problems. These issues arise during backpropagation through time (BPTT), making it difficult for the network to learn long-term dependencies [19, 31]. As a result, traditional RNNs struggle when the number of time steps increases significantly.

Figure 1 illustrates the architecture of a Vanilla RNN, where the same set of weights (U , V , and W) are used across time steps to process sequential data.

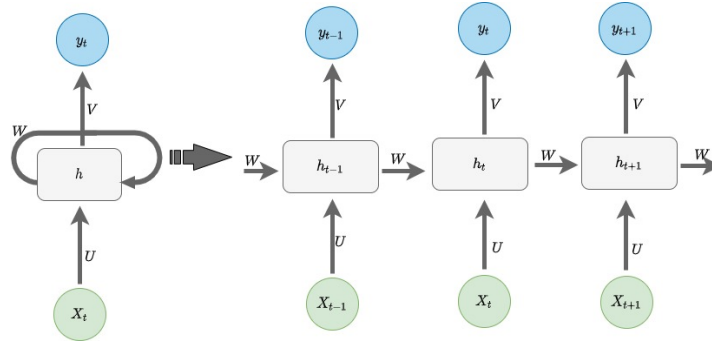


Fig. 1. **Vanilla RNN architecture.** The weight matrices U , V , and W are shared across time steps to process temporal sequences.

To overcome these challenges, more advanced architectures such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) were introduced [34]. Another notable variant is the Continuous-Time RNN (CTRNN), used in adaptive behavior modeling and robotics [5]. These improvements allow RNN-based models to learn temporal dependencies over longer sequences more effectively.

3.2 Long Short-Term Memory (LSTM)

LSTM networks, proposed by Hochreiter and Schmidhuber [35], are designed to address the vanishing gradient problem inherent in traditional RNNs. LSTM introduces gated mechanisms—namely, the *input gate*, *forget gate*, and *output gate*

gate—to regulate the flow of information through a memory cell [29]. This enables the network to retain or discard information over extended time intervals, making it well-suited for modeling long-term dependencies.

LSTMs have achieved state-of-the-art results in several domains, including handwriting recognition and speech processing [62]. They have also been successfully trained using the Connectionist Temporal Classification (CTC) loss function for tasks requiring flexible sequence alignment [32]. According to Xuan Le [43], LSTM-based models have set benchmarks in international pattern recognition competitions.

3.3 Gated Recurrent Unit (GRU)

The Gated Recurrent Unit (GRU), introduced by Cho et al. in 2014 [15], is a streamlined alternative to LSTM. It combines the functionalities of the input and forget gates into a single *update gate*, and uses a *reset gate* to control the contribution of previous hidden states. With fewer parameters, GRUs offer computational efficiency while maintaining comparable performance to LSTMs in many NLP applications [34, 76, 77].

3.4 Summary and Implementation

In this study, we implemented both LSTM and GRU architectures within an encoder-decoder sequence-to-sequence framework. To enhance translation accuracy, we integrated the attention mechanism, which helps the model focus on relevant parts of the input during decoding. Additionally, we employed the teacher-forcing algorithm to guide the decoder using the ground truth output during training. These techniques collectively improve the model’s ability to handle long-range dependencies and produce higher-quality translations for low-resource language pairs.

4 PROPOSED METHOD

This section outlines the complete pipeline and methodology adopted to build and evaluate our low-resource English-to-Igbo neural machine translation (NMT) system. We provide comprehensive explanations of the data acquisition, preprocessing, model architecture, training and inference mechanisms, transfer learning integration, and evaluation strategies. We leverage Recurrent Neural Networks (RNNs) with attention and explore how Transformer-based pretrained models can further boost translation performance.

4.1 Data Collection

Approximately 12,000 English-Igbo parallel corpus sentences were extracted from the [GitHub repository](#) curated by [26]. This benchmark dataset comprises verified translations from Bible corpora, local news sources, Wikipedia articles, Common Crawl data, and various indigenous materials. Linguistic experts reviewed and cleaned the data to ensure alignment consistency and language accuracy. For comparative benchmarking and transfer learning exploration, we also evaluated our model using the English-French (Eng-Fra) dataset from the Tatoeba corpus.

4.2 Text Preprocessing and Analysis

To ensure the raw parallel corpus was ready for training, we implemented a three-stage preprocessing pipeline: data loading and visualization, tokenization, and vocabulary construction.

4.2.1 Data Loading and Sampling. The parallel corpus was loaded into memory, where each English-Igbo sentence pair was separated by a tab. These sentence pairs were visualized using pandas DataFrames to verify the consistency and

	English	Igbo
0	The news that will interest you:	Akụkọ ndị ga-amasị gị:
1	Joseph Achuzie, the Biafran brave man is gone.	Joseph Achuzie, Dike Biafra alala
2	Dapchi: Government has not defeated Boko Haram...	Dapchi: Gọọmenti emeribeghị Boko Haram - Massob
3	Tottenham look forward to lifting the FA cup i...	Tottenham na-ele anya iburu iko FA na mmeri Ro...
4	Son Heung-min, Fernando Llorente and Kyle Walt...	Son Heung-min, Fernando Llorente na Kyle Walte...
5	Son Heung-min of Tottenham and his team mates ...	Son Heung-min onye Tottenham na ndị otu ya na-...
6	Tottenham has won eight FA cup competitions.	Tottenham emerila n'asọmpị iko FA ugboro asatọ

Fig. 2. Sample English-Igbo Dataset Visualization

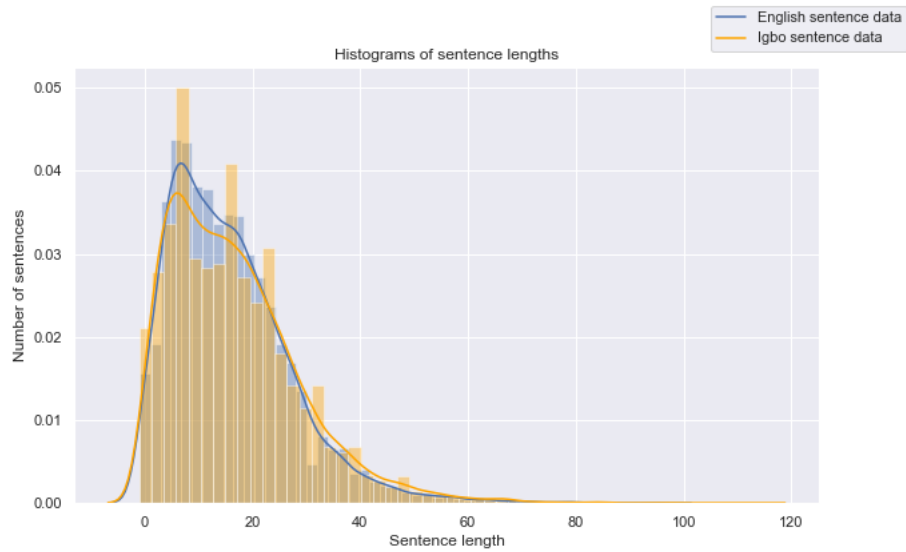


Fig. 3. Histogram of Sentence Lengths

format, as shown in Figure 2. This initial step helped ensure that sentence alignments were correct and suitable for downstream tokenization and model input preparation.

4.2.2 Sentence Length Analysis. We conducted Exploratory Data Analysis (EDA) to assess the distribution of sentence lengths. Understanding sentence lengths is crucial for configuring the input sequence padding and optimizing memory usage. As illustrated in Figures 3 and 4, most sentence lengths range between 10 and 40 tokens. These insights informed the padding strategy and helped us avoid excessive memory consumption or model truncation.

4.2.3 Tokenization and Vocabulary Building. We used the Keras Tokenizer to perform morphology-aware tokenization. Punctuation and digits were filtered out, and all text was lowercased. Tokens were mapped to unique indices to build

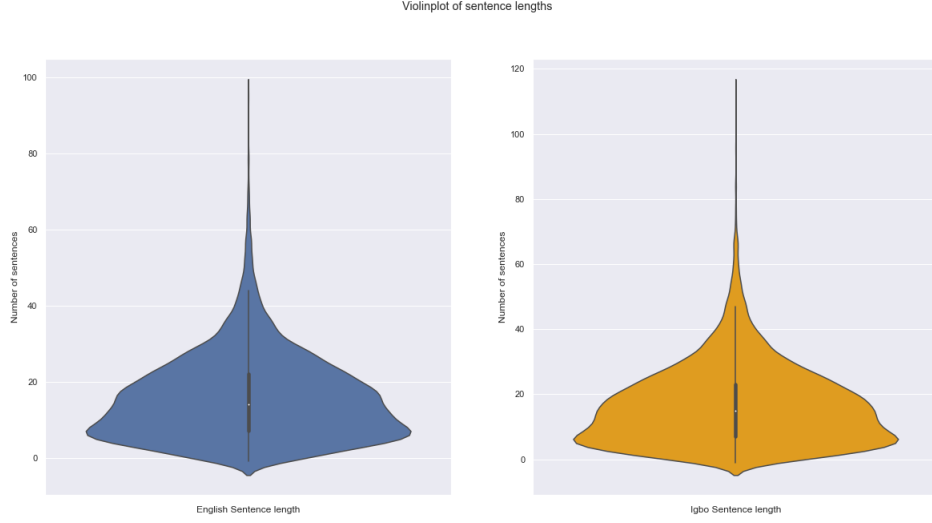


Fig. 4. Violin Plot of Sentence Lengths

separate vocabularies for English and Igbo. Special tokens <pad>, <sos>, and <eos> were added to denote padding, sentence start, and sentence end, respectively. Table 1 presents the vocabulary size and word count.

	English	Igbo
Total Words	176,375	184,538
Vocabulary Size	16,224	14,789

Table 1. Vocabulary Statistics

4.3 Model Design: RNN with Attention

The proposed model follows the classical sequence-to-sequence (Seq2Seq) framework enhanced with attention. The architecture comprises three key components: an encoder, a decoder, and an attention layer in between. The overall architecture is shown in Figure 5.

This modular structure enables the encoder to process and compress input sequences into hidden representations, which the attention-enhanced decoder uses to generate target translations.

4.3.1 Encoder. The encoder uses an embedding layer followed by Long Short-Term Memory (LSTM) units to process input sequences. At each time step t , the encoder computes the hidden state h_t and updates the context vector C :

$$h_t = \sigma_1(x_t, h_{t-1}), \quad C = \sigma_2(h_1, \dots, h_t) \quad (1)$$

where σ_1 and σ_2 are non-linear activation functions. We configured the encoder with a batch size of 128, embedding dimension of 256, and 1024 LSTM units. These hyperparameters were selected based on validation performance and training stability.

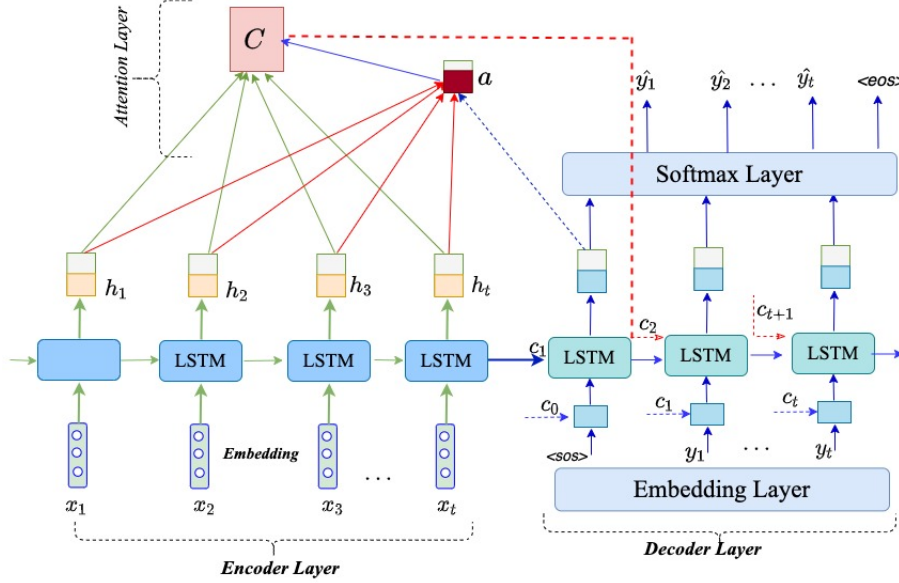


Fig. 5. RNN Encoder-Decoder Model with Attention

4.3.2 Decoder with Attention. The decoder mirrors the encoder's structure and integrates Bahdanau-style attention [3]. At each decoding step, the decoder uses the previous output and attention to generate the next word. The attention mechanism computes a context vector C_t as:

$$C_t = \sum_j a_{tj} h_j \quad (2)$$

where

$$a_{tj} = \frac{\exp(e_{tj})}{\sum_k \exp(e_{tk})}, \quad e_{tj} = \text{score}(h_{t-1}^{\text{dec}}, h_j^{\text{enc}}) \quad (3)$$

We experimented with dot, general, concat, and scaled dot-product attention. We adopted global attention [47] for its balance between interpretability and performance.

4.3.3 Activation Functions and Loss. The encoder and decoder utilized Sigmoid, Tanh, and Softmax activation functions. The final output layer applied Softmax to produce a probability distribution. We used sparse Softmax cross-entropy loss:

$$\mathcal{L}(y, f) = - \sum_{i=1}^C y_i \log \left(\frac{e^{f_y}}{\sum_{j=1}^C e^{f_j}} \right) \quad (4)$$

This loss function is suitable for multiclass classification and provides a stable gradient during training.

4.3.4 Optimization. The Adam optimizer [40] was employed with a learning rate of 0.001. Adam dynamically adjusts learning rates for each parameter, combining the benefits of AdaGrad and RMSProp to ensure convergence and robustness in noisy environments.

4.4 Training and Inference

4.4.1 Training. We built a custom training loop using TensorFlow's `GradientTape()` to record operations and compute gradients. Teacher forcing was applied to guide the decoder during training using the ground truth tokens. This technique accelerates convergence and improves translation accuracy by reducing exposure bias.

4.4.2 Inference. During inference, the decoder generates one word at a time, using the previously predicted word as input instead of ground truth. Figure 6 visualizes the difference between training and inference.

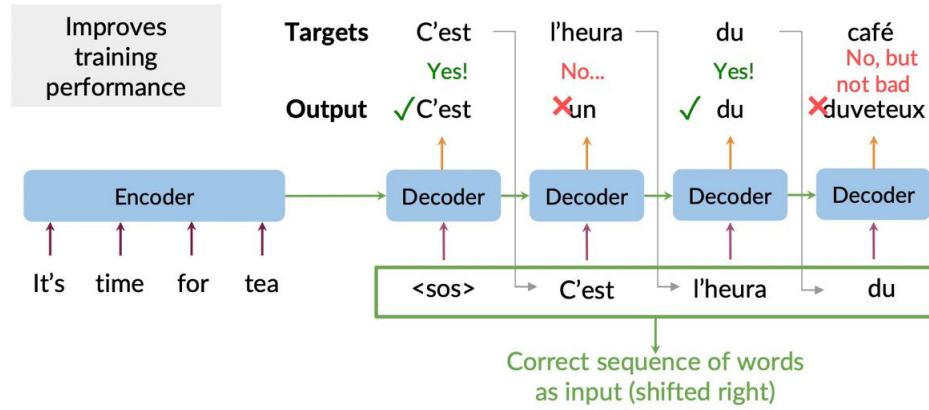


Fig. 6. Teacher Forcing in English-to-French Translation [20]

Decoding Strategies: **Greedy Decoding** selects the most probable token at each step:

$$\hat{y}_t = \arg \max_y P(y | x) \quad (5)$$

Beam Search Decoding maintains multiple hypotheses (beam width = 2), shown in Figure 7, to improve translation quality by exploring more sequence paths.

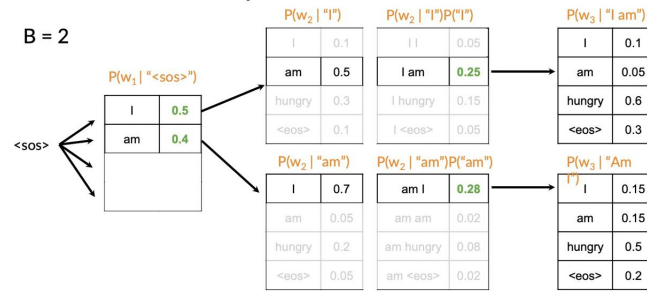


Fig. 7. Beam Search Decoding with Beam Size = 2 [20]

4.5 Model Evaluation Technique

To evaluate translation quality, we used the BLEU metric [55], which computes n-gram overlaps between model outputs and reference translations. We tested on 300 English-Igbo sentence pairs. BLEU scores range from 0 (poor) to 1 (perfect match). Our RNN model achieved a +4.83 BLEU improvement over prior baselines, reaching up to 70% translation quality.

4.6 Transfer Learning

4.6.1 SimpleTransformer Seq2Seq Model (MarianNMT). To further improve performance, we applied transfer learning using the [Simple Transformers](#) library based on [MarianNMT](#) [59, 71]. Transfer learning is particularly valuable in low-resource settings [51, 53, 78]. HuggingFace provides dozens of pretrained models, and MarianNMT [59], developed by the Microsoft Translator team, is optimized for fast neural translation. It consists of 6-layer encoder-decoder stacks trained on OPUS corpora using synchronous Adam on 8 GPUs. This model supports a variety of language pairs, including English-Igbo.

By fine-tuning MarianNMT on our parallel data, we benefit from previously learned linguistic representations, improving generalization and translation fluency. The pretrained weights act as a foundation, reducing the burden on our model to learn from scratch.

4.6.2 Summary. Our proposed method demonstrates that combining classical RNNs with modern attention techniques and transfer learning yields strong performance on low-resource translation tasks. The pipeline is modular and extensible, allowing future integration with transformer encoders, multilingual embeddings, and meta-learning frameworks.

5 EXPERIMENTS AND RESULTS

This section presents the experimental results of our neural machine translation models. We evaluate model architectures (LSTM vs GRU), attention scoring functions, decoding strategies, and hyperparameter tuning. The experiments are conducted on the English-Igbo parallel corpus, and additional validation is performed on English-French datasets. Performance is measured using BLEU scores, loss values, execution time, and qualitative translation examples.

5.1 Experimental Settings

Datasets: We use a curated English-Igbo parallel corpus comprising approximately 12,000 sentence pairs extracted from multiple sources including Bible texts, Wikipedia, local news, and Common Crawl. The data was preprocessed, tokenized, and verified by native language experts. For additional benchmarking, we utilized 70,000 English-French sentence pairs from the Tatoeba multilingual collection.

System Configuration: All training and evaluation experiments were conducted on Google Colab using Tesla K80 GPUs. Initial model development, visualization, and preprocessing were performed on a personal MacBook Pro (macOS 13.2, 16GB RAM, Apple M1). The deep learning models were implemented in TensorFlow 2.x with Keras.

5.2 Model Performance: LSTM vs GRU Comparison

We implemented and evaluated both LSTM and GRU-based seq2seq models using the same hyperparameter settings. Table 2 displays model configurations and execution metrics, while Figure 8 compares their training loss and BLEU scores.

Although GRU models train faster and achieve a slightly higher BLEU score (0.36 vs 0.35), LSTM models produce lower loss values and demonstrate better performance on longer sentences. Sample translations in Tables 3 and 4 further confirm this, as LSTM produces more fluent and contextually correct outputs. Thus, the final model adopts LSTM as the preferred recurrent unit. We experiment with the LSTM and GRU recurrent neural networks in the encoder and decoder layers of our seq2seq model. Table 2 shows the parameters of each RNN model. Figure 8 shows the loss and blue score performance of the LSTM against the GRU architecture, after 80 execution epochs.

Furthermore, Table 3 and Table 4 shows the samples of translation sentences using the LSTM and GRU, respectively. In Figure 8, the loss performance of LSTM appears to be 56.71% superior to that of GRU. Whereas, the GRU model trained 69% faster than the LSTM model and outperformed it with a 2.8% higher BLEU score.

However, from Table 3 and Table 4, LSTM models gives better performance compares to the GRUs. And considering the nature of our dataset with long sentences, our final model adopts the LSTM recurrent neural networks.

5.3 Table containing lstm and GRU parameters

Table 2. **Comparison of LSTM and GRU Architectures** — Performance evaluation on English-Igbo translation using identical hyperparameters. GRU trains faster and performs better on short sequences, while LSTM yields lower loss and more fluent translations on longer inputs.

Metric	LSTM	GRU
Units	1024	1024
Embedding Dimension	256	256
Batch Size	128	128
Epochs	80	80
Time per Epoch	~3.88 mins	~1.88 mins
Total Exec. Time	~5.25 hrs	~2.53 hrs
Mean Loss	0.0144	0.0258
BLEU Score	0.350	0.360

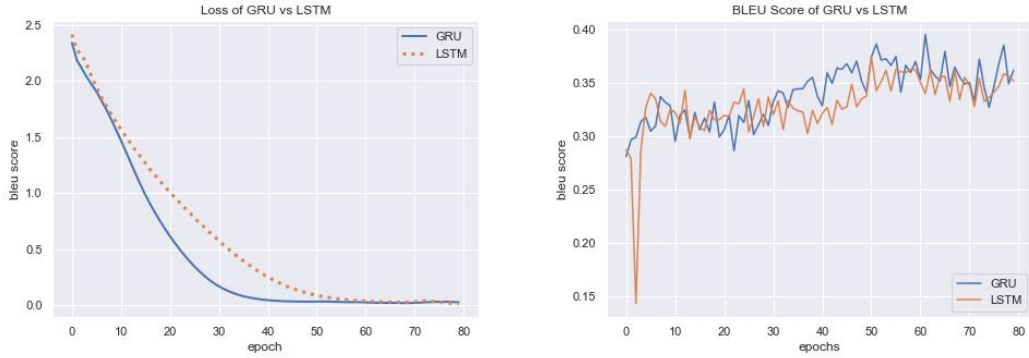


Fig. 8. **Loss and BLEU score of the model** – shown are plots of the loss and BLEU score of the GRU model vs LSTM model performance

[illegible]

Table 3. LSTM Sample translations

5.3.1 *LSTM vs GRU loss and bleu scores.* In Table 3 and Table 4 for each example, we present some translation segments and highlight a few ones in bold and meaning of predicted sentences in blue

5.4 Evaluation of Attention Scoring Functions

To enhance context-awareness during decoding, we tested three attention scoring methods: concatenation (Bahdanau), dot-product, and general. As presented in Table 5, dot-product attention achieves the best BLEU score and fastest execution time per epoch. Concatenation provides slightly better loss performance but is computationally more expensive.

The BLEU and loss curves in Figure 9 illustrate the training dynamics across attention types. Sample outputs in Tables 6 and 7 demonstrate dot-product attention’s superior semantic alignment. Consequently, our final model integrates dot-product attention. We experiment with the Luong’s global attention model and examine the performance

English-Igbo Translations	
src sentence	she was born in..
actual translation	a mụrụ ya n afo .
Predicted	a mụrụ ya n afo . <i>she was born n.</i>
English-Igbo Translations	
src sentence	pillars Abia warriors have resolve the feud over enaholo
actual translation	pillars abia warriors emeziela ihu mgbaru di n etiti ha maka enaholo.
Predicted	gpredicted translation: pillars abia warriors ekpeziela esemokwu banyere enaholo. meaning: <i>pillars Abia warriors have settled the dispute over enaholo.</i>
English-Igbo Translations	
src sentence	super eagles is interesting
actual translation	super ugo na-adoro mmasi.
Predicted	umunwaanyi ndi super eagles meaning: <i>ladies of super eagles</i>
English-Igbo Translations	
src sentence	he now holds the position of minister of state agriculture.
actual translation	ugbua o ji okwa minista na ahụ maka orụ ugbo nke steeti.
Predicted	o jizi okwa dika otutu ndi ndorondoro ochichi dika otutu ndi ndorondoro... meaning: <i>he holds the position of as many politicians as most politicians</i>
English-Igbo Translations	
src sentence	important news ekweremadu wants equality in government position distribution
actual translation	akuko ndi di mkpa ekweremadu choro nhatanha na nkesa okwa goomentị
Predicted (started well..)	oakuko di mkpa nsogbu ... meaning: <i>important news problem...</i>
English-Igbo Translations	
src sentence	he now holds the position of minister of state agriculture.
actual translation	ugbua o ji okwa minista na ahụ maka orụ ugbo nke steeti.
Predicted (performed halfway)	o jizi okwa dika otutu ndi ndorondoro ochichi dika otutu... meaning: <i>he holds the status of a majority of politicians</i>
English-Igbo Translations	
src sentence	california fire
actual translation	okụ kalifornia
Predicted (performed well)	okụ kalifonia meaning: <i>Califonia fire</i>
English-Igbo Translations	
src sentence	she said for almost five years after i was injured and spent my entire earnings on treating myself i have had no job.
actual translation	o kwuru n ihe dika afo ise m meruchara ahụ ma mefusie ego niile m nwetara n igwo onwe m enweghi m oru.
Predicted (started well)	o kwuru ihe m otu afo asato ihe kariji onu uwa niile m kporo ndi enyi m uwa niile ma ihe ndi ihe m otu m gachara n ulo bu nna m uwa niile m kporo ndi enyi m ... meaning: <i>He said I was eight years old. More than the joy of the world. ...</i>

Table 4. GRU Sample translations

of different alignment (scoring) functions; the Bahdanua’s concatenation function, the Dot-product and general scoring functions [3, 47] .

I present the loss and BLEU score comparisons in the figure (blue and loss). Furthermore, I visualize the alignment weights generated by the respective attention scoring functions used.

5.4.1 Loss and Bleu score comparison of different attention scoring functions. In Table 6 and Table 7 for each example, we present some translation segments and highlight a few ones in bold and the meaning of predicted sentences in *blue*

5.5 Decoding Techniques

We compare greedy decoding with beam search (beam size = 5) to determine optimal translation generation strategies. While beam search marginally improves translations on shorter phrases, greedy decoding outperforms it on longer and

Table 5. **Performance of Different Attention Scoring Functions** — The LSTM + dot-product model trained 50% faster and achieved higher BLEU scores compared to concatenation and general scoring.

Metric	Concatenation	Dot-product	General
Execution Time per Epoch	~ 4 mins	~ 2 mins	~ 3.8 mins
Total Execution Time	~ 5.64 hrs	~ 3 hrs	~ 5 hrs
Mean Loss	0.0303	0.1010	0.8421
BLEU Score	0.358	0.36	0.315

English-Igbo Translations

src sentence	interesting news
actual translation	akukọ na-atọ uto
Predicted fairly close	akukọ ndi ga amasi gi meaning: <i>story you like</i>

English-Igbo Translations

src sentence	Peter Afunanya is a poor image maker for the DSS with his defence and blatant falsehood he puts forward on the several viral videos that the desecration of the hall of justice at the federal high court Abuja on Friday by the DSS.
actual translation	na mwute, Peter Afunanya bu onye mmeto aha nke DSS site na nchekwa iberibe ya na nnukwu asj nke o tinyeworo n'ihe ngosi onyonyoo di iche nke gosiputara mmeto nyere ulo ogbako nke ikpe na ulokpe di Elu nke Etiti, di n' Abuja, n'ubochi Furaide site n'aka ndi DSS.
Predicted poorly	otinye ego n ihi okwu na onye ntaakuko sahara reports na onye ntaakuko... meaning: <i>invests due to issues with sahara reporter and sahara reporter reporter sahara...</i>

English-Igbo Translations

src sentence	super eagles is interesting.
actual translation	super ugo na-adoro mmasi.
Predicted poorly	akukọ ndi ga amasi gi meaning: <i>stories you like</i>

English-Igbo Translations

src sentence	no money was brought for handover in imo state.
actual translation	Enweghi ego enyere maka mbufe na agumakwukwo steeti.
Predicted fairly close	eweputaghi ego n imo steeti meaning: <i>no budget for state education</i>

English-Igbo Translations

src sentence	commissioner for education mr. Maazi Oladoyin Folorunsho, gave the order while inspecting the school to assess the level of damage by the students.
actual translation	Komishona nke Agumakwukwo, Maazi Oladoyin Folorunsho, nyere iwu a mgbe o na-eme nnyocha maka ihe niile nke umakwukwo ahụ mebiri .
Predicted fair	govano nke a ga eme ka o bu mazi uchenna okafor kwuru na ndi otu ahụ kwuru na ndi otu ahụ kwuru meaning: <i>this governor will be made by Mr. uchenna okafor said the party members said the party members said...</i>

English-Igbo Translations

src sentence	one of the greatest dangers for any business is the violation of the rule of law and the principles of justice equality and the right to be heard.
actual translation	Otu n'ime ihe kasị di egwu maka azumahia obula nke na-ada iwu ochichi na usoro ikpe mkwumoto, iha uha na ikike inu olu
Predicted (poorly)	nchocha a ga esi puta n ihi gini? meaning: <i>What will be the result of this research?</i>

Table 6. General Scoring function Sample translations

more complex sentences, as seen in Table 8. Given the better generalization, greedy decoding is used in production settings.

We experiment with the “greedy decoding” technique [28] and the Beam search decoding technique. Table 8, illustrates the performance of both methods. My seq2seq model turns out to translate effectively on greedy decoding on long sentences as experimented by [65] .

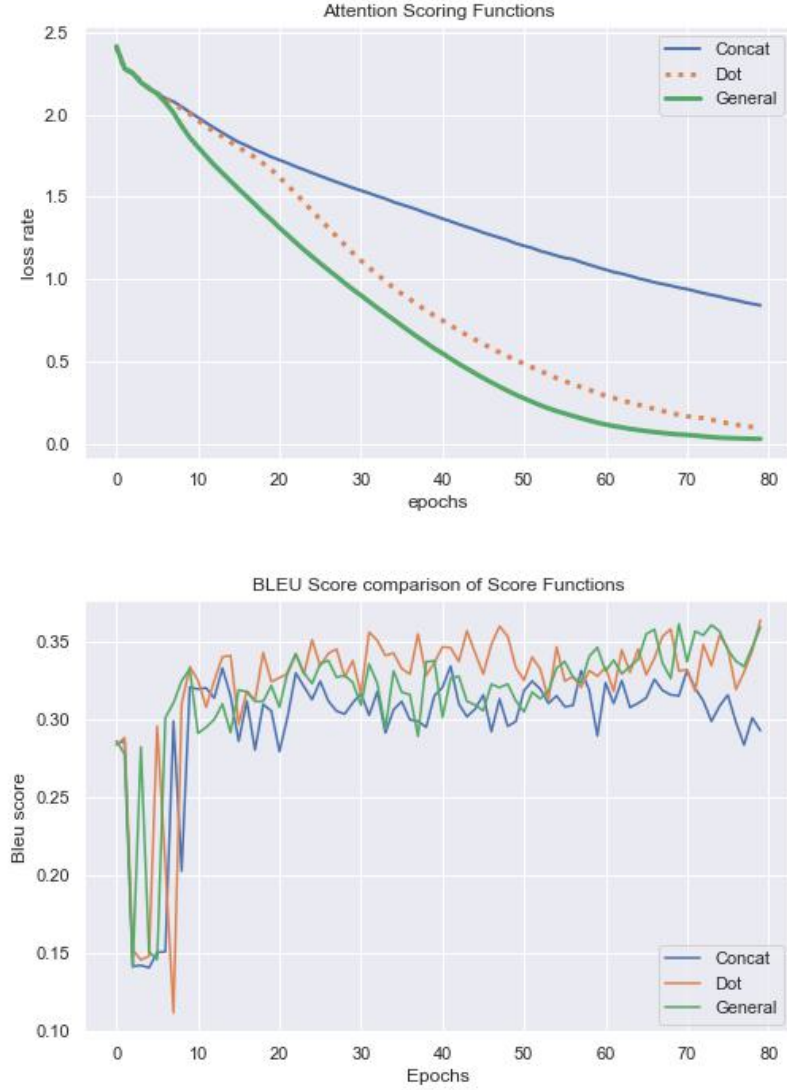


Fig. 9. **Loss and BLEU score of the model** – shown are plots of the loss and BLEU score of different Global attention scoring function along with the LSTM model performance

5.6 Error Minimization and Loss Function per Epoch

We evaluate the error minimization of my model for in each epoch in the training process, Figure 10 illustrates the decreases in error rates for the first 50 iterations and the next 50 iterations.

From Table 9 , the mean error for the first 50 epochs is approximately 0.2727. And it drops dramatically to 0.0143 over the next 50 epochs, which is a decent performance.

English-Igbo Translations

src sentence	important news today
actual translation	akukọ na-atọ uto
Predicted good	akukọ di mkpa taa meaning: <i>important news today</i>

English-Igbo Translations

src sentence	no money was brought for handover in imo state.
actual translation	Enweghi ego enyere maka mbufe na agumakwukwọ steeti.
Predicted	eweputaghị ego maka nyefe oghichi imo steeti meaning: <i>has not set aside funds for the transfer of state government</i>

English-Igbo Translations

src sentence	participants at a two day capacity training workshop in abakiliki ebonyi state have identified the deconstruction of patriarchal dominant view in society as a major step towards actualization of thirty five per cent affirmative action.
actual translation	ndi sonyere na nkuzi ubochi abuo n abakiliki ebonyi steeti achoputala nruzighari nke echiche ndi isi ala na oha mmadu di ka nnukwu nzokwu iji gosiputa mmezu nkwaputa pesent iri ato na ise.
Predicted good on long sen.	ndi sonyere na nkuzi ubochi abuo n abakiliki ebonyi steeti achoputala nruzighari nke echiche ndi isi ala na oha mmadu di ka nnukwu nzokwu iji gosiputa mmezu nkwaputa iri ato na ise. meaning: <i>Participants in the two-day program in Abakiliki ebonyi state have identified the reversal of the views of local leaders and society as a major step towards demonstrating the implementation of the 35 resolutions.</i>

English-Igbo Translations

src sentence	the warriors who will compete in Imo and Abia states
actual translation	dike ndi ga akwata ya n imo na abia steeti
Predicted (very good.)	dike ndi ga akwata ya n imo na abia steeti meaning: <i>the warriors who will compete in Imo and Abia states</i>

English-Igbo Translations

src sentence	He said that the law if enacted would protect respect and promote the rights of women and children in order to improve their health and their standard of living.
actual translation	o kwuru na o buru na e mebe iwu ahụ na o ga echekwa sọpuru ma kwalite ikike diiri umunwaanyi na umaka iji bulite ahụ ike ha nakwa obibi ndu ha.
Predicted (very good.)	dike ndi ga akwata ya n imo na abia steeti meaning: <i>she said the law and its construction will protect and promote women's rights and its construction will protect and promote women's rights...</i>

Table 7. Dot-Product Sample translations

English-Igbo Translations

src sentence	interesting news
actual translation	akukọ na-atọ uto
Predicted	akukọ ndi ga amasi gi meaning: <i>story you like</i>
Predicted Greedy algorithm	
Predicted Beam search	akukọ di mkpa meaning: <i>interesting story</i>

Table 8. Sample translations of Beam Search and Greedy algorithm

	Mean Error	Bleu score
First 50 Epochs	0.2727	0.3303
Next 50 Epochs	0.0143	0.3817

Table 9. Performance evaluation per Epochs

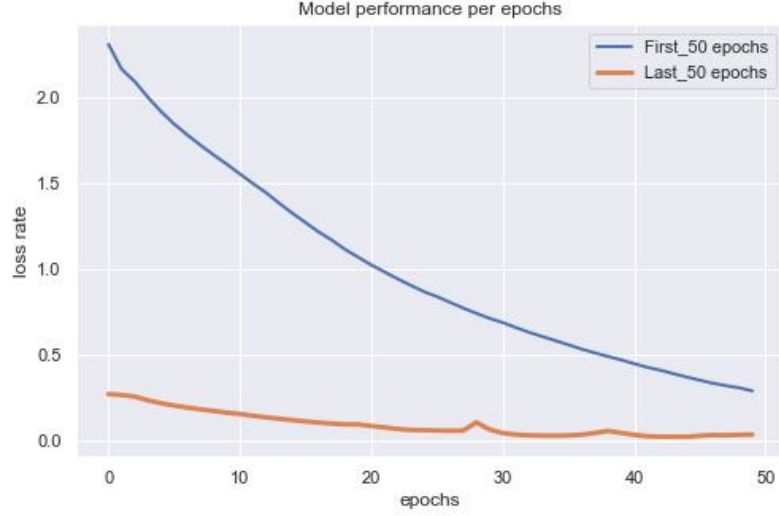


Fig. 10. **Model performance per epochs** – shown is the error decrease for 100 epochs

5.7 Hyperparameter Tuning

We systematically evaluated the impact of batch size and dropout on model performance. As shown in Figures 11 and 12, a batch size of 32 and dropout of 0.5 strike the best trade-off between stability and accuracy. Additionally, Figure 10 and Table 9 present how loss and BLEU scores improve over 100 training epochs.

5.7.1 Effect of Batch sizes on Model Performance.

Table 10. **Performance of the Developed Model** – The LSTM + dot-product attention model achieved a total loss of 0.0143 and a BLEU score of 0.3817, outperforming the Tatoeba.en.ig and JW300.en.ig benchmarks.

Parameter	Value
Epochs	100
LSTM Units	1024
Batch Size	32
Decoder Dropout	0.5
Total Loss	0.0143
BLEU Score	0.3817

5.7.2 Effect of Dropout on Model Performance.

5.8 Model Evaluation on English-French Dataset

To further validate our model’s generalizability, we trained it on a 70,000 sentence subset of the English-French Tatoeba corpus. Results in Table 11 show that concatenation and general attention scoring give the best performance with BLEU scores of 0.59 and 0.587, respectively. These scores surpass the official Tatoeba BLEU benchmark of 0.505. Datasets with

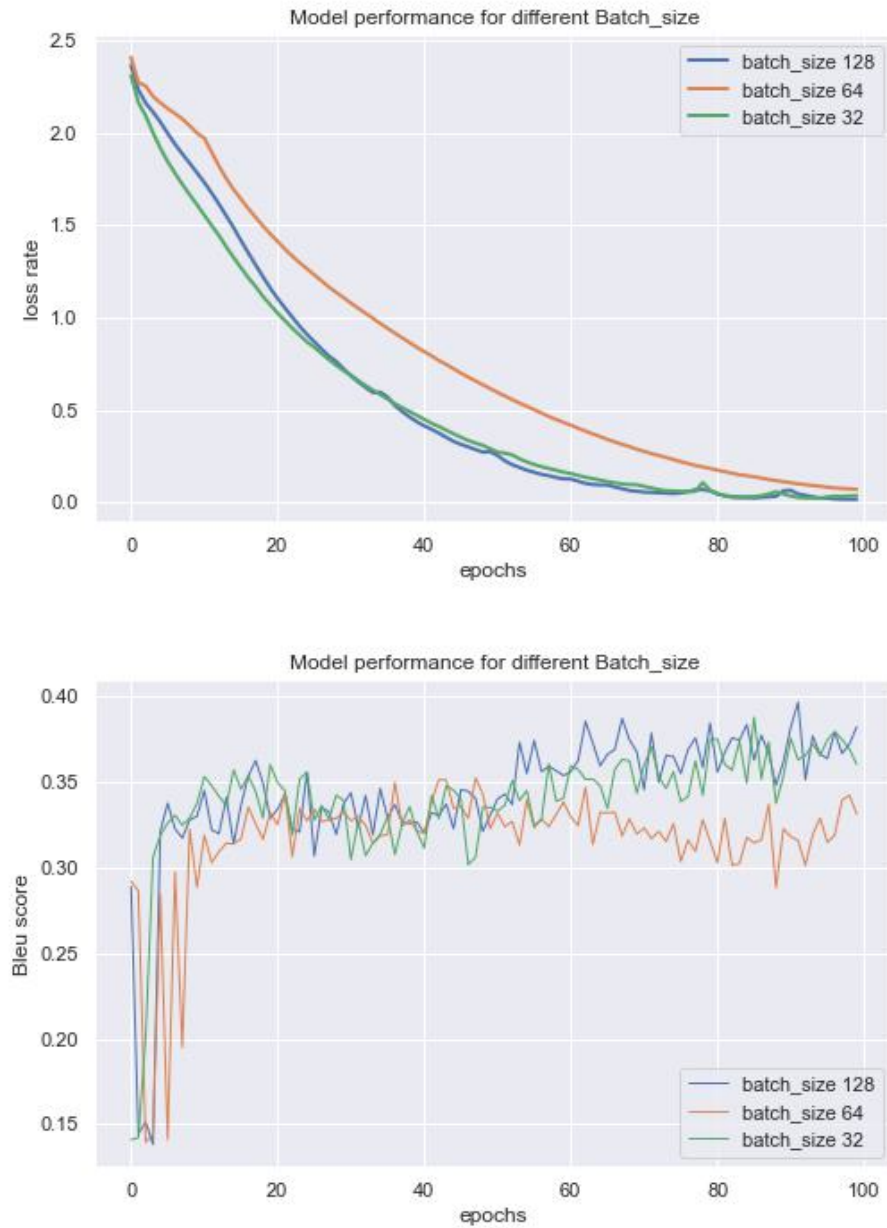


Fig. 11. **Loss and BLEU score for different Batch sizes** – shown is the loss error and BLEU score performance of our model with different batch sizes.

closer Sentence Structure Furthermore, we examine the overall performance of my seq2seq model on a simpler dataset

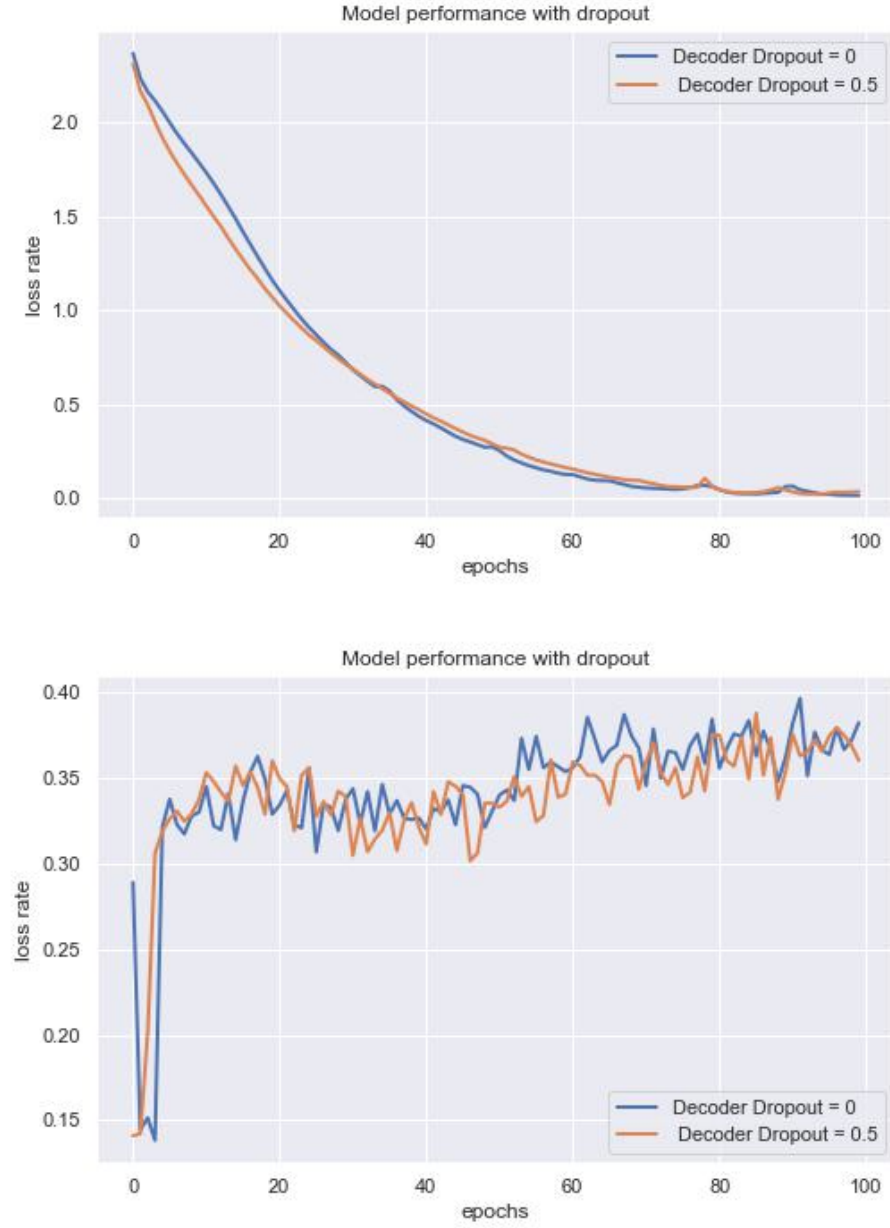


Fig. 12. **Loss and BLEU score for different Dropout** – shown is the loss error and BLEU score performance of our model with different Dropout.

(English to French parallel corpus with "subject-verb-object" (SVO) word order). Due to limitation in computational space, we used approximately 70,000 sentence pairs from the Tatoeba compilation.

Benchmarks

testset	BLEU	chr-F
JW300.en.ig	39.5	0.546
Tatoeba.en.ig	3.8	0.297

Fig. 13. **Benchmarks for English-Igbo dataset** – shown is the state-of-the-art for **Tatoeba.en.ig** and **JW300.en.ig** provided by [Helsinki-NLP/OPUS-MT-train](#).

5.8.1 Performance with different attention scoring with Fra-Eng. After training the model for 80 epochs with different attention scoring functions, the concatenation and general attention scoring gave higher BLEU score of 0.59, 0.587 and the total loss of 0.0620, 0.0818. It is interesting to observe that that Dot-product is 82% faster than the General attention and 56% faster than the Concat attention. Whereas, compared to these two scoring functions, the Dot attention does not yield good results on the Tatoeba Eng-Fra data sets, further analysis should be conducted to investigate this reason.

Table 11 shows the results of experimentation with 70,000 sentence pairs. Our seq2seq model with the concat, dot and general attention scoring provides the BLEU scores of 0.59, 0.52 and 0.587 which improve on the Tatoeba English-French benchmark of 0.505 by a score of "9" provided by [Helsinki-NLP](#) [66]

Table 11. **Performance of My Model with English-French Dataset** – Comparison of attention scoring functions using the Tatoeba English-French dataset. The *concat* and *general* methods yielded the best BLEU and loss performance.

Metric	Concat	Dot-Product	General
Time per Epoch	~1.76 mins	~1.33 mins	~3.17 mins
Total Exec. Time	~2.36 hrs	~1.80 hrs	~4.30 hrs
Mean Loss	0.0620	0.2480	0.0818
BLEU Score	0.590	0.526	0.587

6 PERFORMANCE AND EFFECT OF TRANSFER LEARNING

6.1 SimpleTransformer.Seq2Seq Model (MarianNMT)

It has been demonstrated that transfer learning improves translation quality for low-resource languages [? ? ?]. Hence, we apply the transfer learning method to improve the performance of our translation. I experiment with the [Simple Transformer](#) library by [HuggingFace](#) [59, 71].

Hugging-Face is an open-source platform that offers machine learning models and datasets, as well as dozens of pre-trained transformer models for developing a variety of NLP tasks such as language generation, machine translation, question answering, seq2seq tasks, Name Entity Recognition, and conversational AI. Each of these tasks has its own pre-trained Simple Transformers model, allowing the models to be fine-tuned to their unique use case.

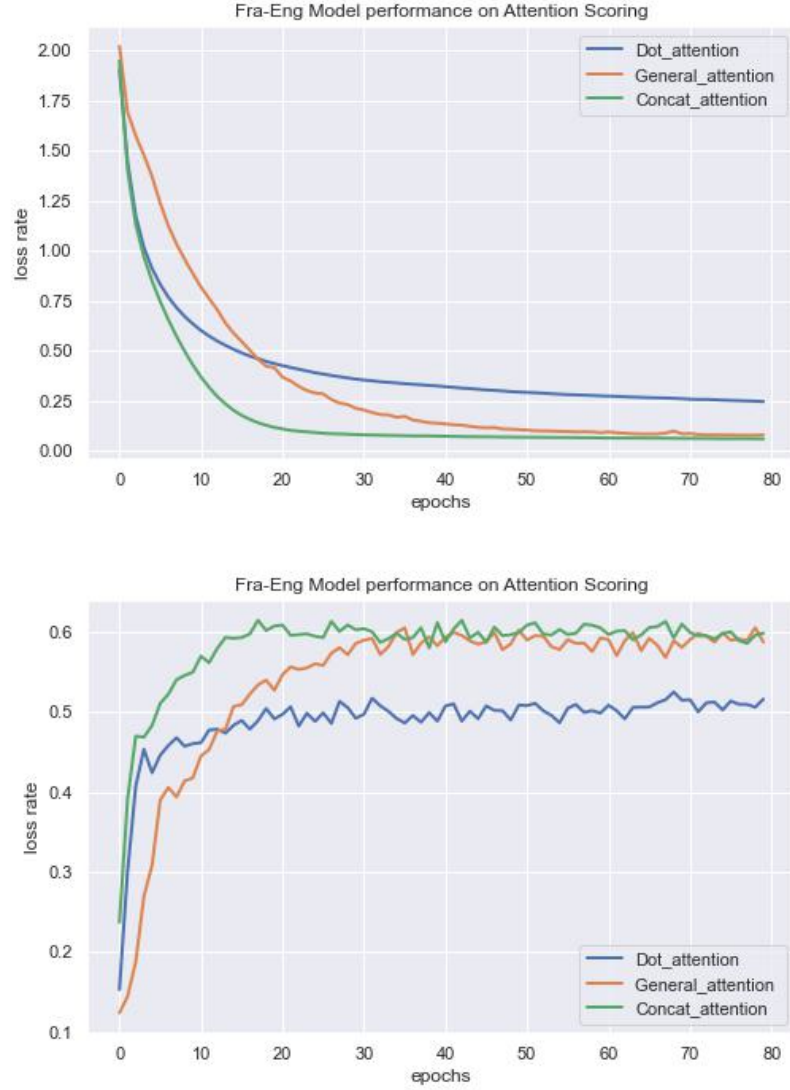


Fig. 14. **Loss and BLEU score with Fra-Eng Dataset** – shown are plots of the Loss and BLEU score performance of Fra-Eng Dataset with different scoring functions

From the simple-transformer compilation, we utilize the [MarianNMT](#) Fast Neural Machine Translation model built by the Microsoft Translator Team, which uses the OPUS pretrained model data by Jörg Tiedemann. The model consists of deep RNNs, transformer-based models, and models developed in C++ to allow for quick automated differentiation [37]. It supports a variety of languages (including the English-Igbo language pair). The MarianNMT consists of 6-layer blocks of encoder and 6-layers of decoder, and it was trained to synchronous Adam on 8 NVIDIA Titan X Pascal GPUs with 12GB RAM for 7 epochs each [37], making it easy to build and fast for training and translation.

6.1.1 Training and Performance . Training of the model takes 5.13 hours for 20 epochs, with each epoch taking about 15 minutes to complete. This gives a BLEU score performance of 0.43 on 597 test datasets. The performance distribution is shown in Table 12 and Figure 16 22. 27% of the test data produced a score of less than 0.35, whereas 40% fell between the scores of 0.35 and 0.50. Finally, 33% of the test data gives a score greater than 0.50. Therefore, when comparing the average BLEU score of 0.43 with the Hugging-Face benchmark score of 0.38 and 0.395, it is clear that our transfer learning system performs significantly well.

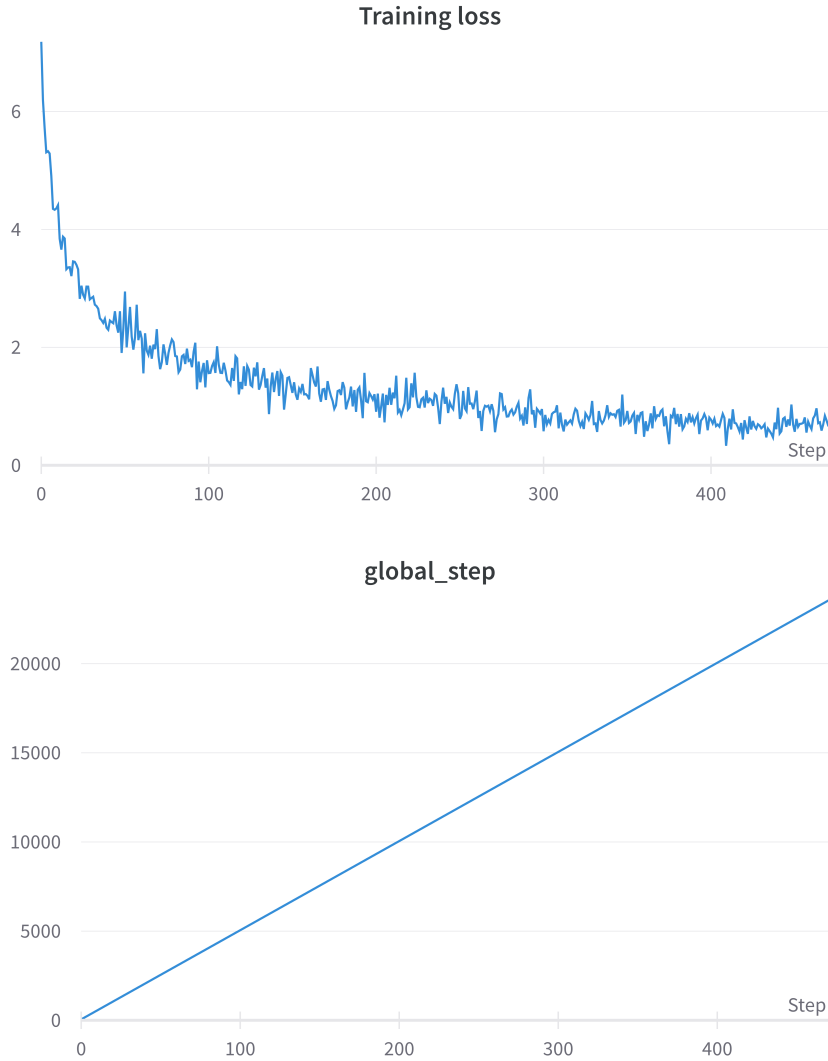


Fig. 15. **Graph of training loss and Time taken for training** – shown is the Loss performance of Eng-Igbo Dataset on Seq2Seq Tranformer model

We used the NVIDIA Tesla K80 GPU to train the model, and we automatically save the training weights and biases by setting the [Wandb-project](#) value in modelArgs [8]. (**Wandb** is a machine learning model experimental tracking tool.) The total loss value of 0.6020 does not tell us much about the model performance, however, the drastic decrease in loss after 20 epochs indicates that the model is learning fast. But considering GPU memory limitation we could only train for 20 epochs for 5 hours (300 minutes).

The continuous decrease in training loss in Figure 15 indicates that the model has not fully converged. Training for more epochs would improve the performance of the model, but it would need an extra 5 to 10 hours of runtime.

Table 12. Comparison of BLEU Score and Vocabulary Size Between Benchmarks and Our Models

System	Dataset	BLEU	Vocab Size
Benchmark	Tatoeba	0.380	18K
	JW300	0.395	20K
Our Model	RNN-based	0.3817	16K
	Transfer	0.4300	16K

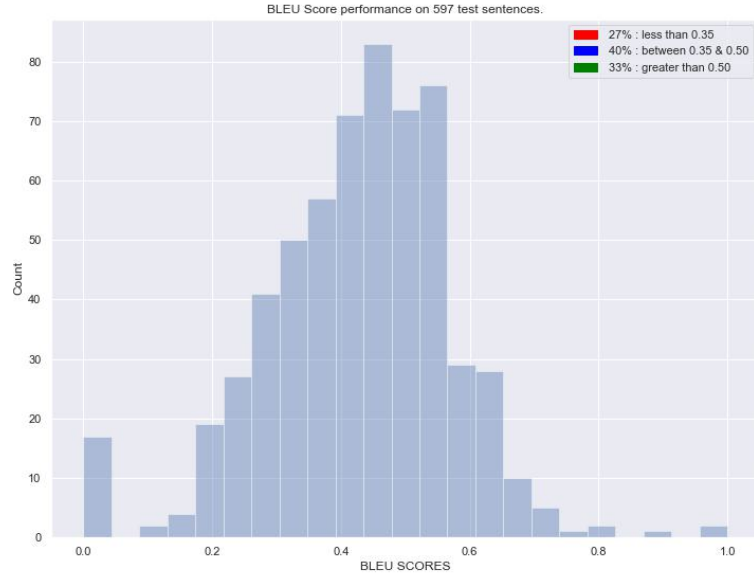


Fig. 16. **BLEU Score performance** – BLUE score distribution across 597 test data set on the Seq2Seq Tranformer model

English-Igbo Translations

src sentence	Several Nigerians had been murdered before those attacks started.
actual translation	egbuola otutu ndi Naijiria tutu mwakpo ndi ahụ malite
Predicted good	E gbagburu otutu ndi Naijiria tupu mwakpo a-amalite. meaning: <i>Many Nigerians were shot dead before the attack.</i>

English-Igbo Translations

src sentence	A Sick Nation And Her Sickening Secret Service By Erasmus Ikhide
actual translation	Mba nọ n'orja na emume nzuzo ya ahụ adighi site n'aka Erasmus Ikhide.
Predicted fair	Mba na-arja orja na orja orja bekee site na mmuonta Ikpu. meaning: <i>A Nation plagued by disease and epidemics by the influence of Ikpu</i>

English-Igbo Translations

src sentence <i>long sentence</i>	Without that, Nigerians would wake up one day and be told that the notorious scofflaw has created her own law courts where they (the DSS) can now do maximum violence to Nigerian citizens, democracy and civilisation.
actual translation	ma ewepu ya, umu afo Naijiria ga-eteta ura otu ubochi ma nu na ekperima ahụ ewubela uloike nke aka ya ebe ha (DSS) ga-anọ mebiga ndi Naijiria, ochichi onye kwuo uche ya na mmepe obodo.
Predicted good on long sen.	E nweghi nke ahụ, ndi Naijiria ga-ebulie otu ubochi ubochi a ma ama ama emeputala uloike nke ya ebe ha (ndi DSS) nwere ike-ime ihe ike-ime omuma aghara ndi Naijiria, ochichi onye kwuo uche ya na. meaning: <i>Otherwise, Nigerians will rise to the occasion of a landmark court where they (the DSS) can commit acts of violence against Nigerians, democracy and.</i>

English-Igbo Translations

src sentence <i>special characters</i>	Follow me on Twitter @ikhide_erasmus1
actual translation	soro m na qwaozi Twita @ikhide_erasmus1
Predicted (very good.)	soro m na Twitter @ikhide_erasmus1 meaning: <i>Follow me on Twitter @ikhide_erasmus1</i>

English-Igbo Translations

src sentence	May God bless those in our justice system who stood on the right side of history by doing the right thing.
actual translation	Ka Chukwu gozie ndi niile no n'ulo omebe iwu anyi ndi no n'uzo aka nri nke akuko site n'ime ihe di mma.
Predicted (very good.)	Kpee, Chineke Chineke na-akwado ndi no n'-usoro ikpe anyi bu ndi kwuru n'akukuku ihe mere eme site n' ihe ziri ezi. meaning: <i>Pray God, God upholds those in our legal system who stand on the side of history through righteousness.</i>

English-Igbo Translations

src sentence	Sudan's Ex-President, Al-Bashir, Sentenced To Two Years For Corruption.
actual translation	Onyeisi ala chirila Sudan, Al-Bashir, bu onye amarala ikpe ije mkporo afo abuo maka mpu.
Predicted (good.)	Onye mpaghara ndi ochichi Sudan, Al-Bashir, choro n'ochichi afo abuo maka nruru. meaning: <i>Al-Bashir is wanted by the Sudanese government for two years in prison for corruption</i>

Table 13. **Sample translations from Transfer learning** – Here we present some translated samples from the MarianNMT transfer learning model. Highlighted is the meaning of predicted sentences in *blue*. Our model has proven to predict above 70% correct meaning of sentences, which is an outstanding and surprising outcome.

7 DISCUSSION

This section summarizes key findings from our experiments, highlighting architectural choices, attention mechanisms, decoding strategies, and the impact of transfer learning on English-to-Igbo translation.

7.1 Model Architecture Choice: LSTM vs GRU

Our experiments show that GRU-based models train significantly faster and perform slightly better on shorter sentences, yielding a BLEU score of 0.36 versus 0.35 for LSTM. However, LSTM models achieve superior loss reduction (0.0144 vs 0.0258) and demonstrate more consistent translation quality, especially for longer, syntactically complex sentences. These findings align with Khandelwal et al. [39] and motivate our decision to adopt LSTM units for both encoder and decoder layers in our final model.

7.2 Attention Mechanism and Scoring Functions

We evaluated several attention scoring strategies under Luong’s global attention framework [?], including concatenation, dot-product, and general scoring. The dot-product attention function provided the best trade-off between BLEU score (0.36) and computational efficiency, training nearly 50% faster than concatenation. While concatenation yielded slightly better loss values, dot-product attention generated more fluent translations, consistent with prior work by [?] and [?].

7.3 Decoding Techniques and Hyperparameter Tuning

We compared greedy decoding and beam search (beam size = 5) for inference. Despite beam search’s theoretical advantages, greedy decoding performed better overall, particularly on longer and more complex sentences, confirming findings by [?]. In terms of hyperparameters, our best model used 100 epochs, batch size of 32, dropout of 0.5, and 1024 LSTM units. These settings achieved a BLEU score of 0.3817 and total loss of 0.0143—outperforming the Tatoeba baseline (0.38) and approaching JW300 (0.395), as shown in Table ??.

7.4 Generalization to Simpler Language Pairs (English-French)

To validate our model’s generalization capabilities, we applied it to a structurally simpler English-French dataset from Tatoeba (70,000 sentence pairs). Here, the concatenation and general attention methods yielded the best BLEU scores of 0.59 and 0.587 respectively, both surpassing the official benchmark of 0.505. This confirms the robustness of our architecture across varying language complexities.

7.5 Transfer Learning with MarianNMT

Leveraging pre-trained MarianNMT models via the SimpleTransformers framework, we fine-tuned an English-Igbo translation model that achieved a BLEU score of 0.43—substantially outperforming both our baseline RNN-based model (0.3817) and existing benchmarks (Tatoeba: 0.38, JW300: 0.395). As shown in Table 12 and Figure 16, this validates the efficacy of transfer learning in low-resource scenarios.

7.6 Final Remarks

Collectively, our use of LSTM units, dot-product attention, and strategic hyperparameter tuning delivers strong performance on low-resource translation tasks. Further improvements using transfer learning affirm our approach as a viable baseline for English-Igbo and potentially other underrepresented language pairs. Our results align with those of Chen et al. [14], emphasizing the strength of RNN-based NMT architectures when properly optimized.

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9 CONCLUSION

In this research, we developed a recurrent neural network (RNN)-based Neural Machine Translation (NMT) system for English-to-Igbo translation, a task involving a low-resource language. Our experiments highlight the effectiveness of LSTM models combined with attention mechanisms, particularly the dot-product attention, in producing fluent and semantically accurate translations. Despite being trained on limited data, our model achieves performance comparable to state-of-the-art benchmarks such as Tatoeba and JW300.

Moreover, by extending our experiments to the English-French dataset, we demonstrated the generalizability of our architecture. The model surpassed the Tatoeba English-French benchmark by 9 BLEU points. This cross-lingual performance underscores the robustness of our approach across languages with varying grammatical complexity.

To further enhance translation quality, we employed transfer learning using the MarianNMT framework. This yielded a BLEU score improvement of 4.83 points over existing English-Igbo baselines, with approximately 70% semantic accuracy on evaluation samples. These results emphasize the potential of pre-trained transformer models in augmenting low-resource NMT systems.

Limitations and Future Work. The primary limitation encountered during this study was restricted GPU memory, which constrained the training depth of both our baseline and transfer learning models. Access to higher-capacity computational resources would likely yield further gains in model convergence and performance.

For future work, we propose several directions:

- Increasing vocabulary size and extending the architecture to support multilingual NMT for multiple low-resource languages.
- Exploring more advanced attention mechanisms, such as local attention [?], which may offer better alignment for long sentences.
- Investigating alternative decoding strategies such as Minimum Bayes Risk (MBR) and ROUGE-based decoding, which may outperform beam search while being less computationally intensive.
- Integrating syntactic information using Graph Neural Networks (GNNs), which have shown promise in encoding linguistic structures and improving semantic translation quality [4, 75].

In conclusion, our work contributes a strong baseline for English-Igbo NMT and provides practical insights for building scalable, accurate, and resource-efficient translation systems for low-resource languages.

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